

Week 9: Spectral Estimation

Homework, Weekly Quiz, and Solutions

DSP + ML • Oppenheim & Schaffer / Raschka

Homework Problems

Recommended: assign Problems 1–6 to everyone. Problems 7–8 are useful senior/graduate stretch problems or group extensions.

1. Wiener–Khinchin and power

A zero-mean WSS process has autocorrelation $r_{xx}[m] = (0.6)^{|m|}$. (a) State how its PSD is obtained. (b) Compute the average power. (c) Is its PSD more concentrated at low or high frequency? Explain.

2. PSD through an LTI filter

White noise with variance $\sigma^2 = 2$ drives the FIR $h[n] = \{1, 1\}/2$. Derive $S_{yy}(e^{j\omega})$ and compute the output variance.

3. Periodogram calculation

For $x[n] = \{1, -1, 1, -1\}$, compute the $N = 4$ DFT and periodogram $I[k] = |X[k]|^2/N$. Identify where the power lies.

4. Why the periodogram is inconsistent

For white Gaussian noise, explain why increasing N can improve frequency-grid detail while the periodogram at a fixed frequency still has large relative variability. State the consistency criterion that fails.

5. Bartlett design

A 8192-sample record is sampled at 2048 Hz. Bartlett’s method uses $K = 8$ equal non-overlapping segments. Find the segment length, nominal bin spacing, and qualitative variance change relative to one full-record periodogram. What information is sacrificed?

6. Welch design trade-off

You need to detect two peaks 8 Hz apart in a 2 kHz sampled signal. Compare $n_{\text{perseg}} = 128$ and $n_{\text{perseg}} = 1024$. Which is more plausible for resolving the peaks, and which gives more averaging opportunities for a fixed record length?

7. Window / overlap reasoning

Explain why Hann-window Welch estimates commonly use overlap. Why is “twice as many overlapping segments” not equivalent to “twice as many independent estimates”?

8. ML spectral features

Design a compact PSD feature vector for classifying two machine states whose vibration differs mainly in the 80–120 Hz and 220–280 Hz bands. Include preprocessing, train/test discipline, and one reason Welch features might outperform raw FFT magnitudes.

Homework Solutions

1. Wiener–Khinchin and power

By Wiener–Khinchin, $S_{xx}(e^{j\omega})$ is the DTFT of $r_{xx}[m]$. For $r_{xx}[m] = a^{|m|}$ with $|a| < 1$, the closed form is $(1 - a^2)/(1 - 2a \cos \omega + a^2)$. With $a = 0.6$:

$$S_{xx}(e^{j\omega}) = 0.64 / (1.36 - 1.2 \cos \omega).$$

Average power is $r_{xx}[0] = 1$. The correlation is positive and slowly decays with lag, so low-frequency components are emphasized; the PSD is largest near $\omega = 0$.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

2. PSD through an LTI filter

$$H(e^{j\omega}) = (1 + e^{-j\omega})/2 = e^{-j\omega/2} \cos(\omega/2), \text{ so } |H|^2 = \cos^2(\omega/2).$$

$$S_{yy}(e^{j\omega}) = 2 \cos^2(\omega/2) = 1 + \cos \omega.$$

Output variance = $\sigma^2 \sum |h[n]|^2 = 2(1/4 + 1/4) = 1$. Equivalently, it is the area under S_{yy} divided by 2π .

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

3. Periodogram calculation

The sequence is $x[n] = (-1)^n$, exactly the DFT bin at $k = 2$. The 4-point DFT is $X = \{0, 0, 4, 0\}$.

$$I[k] = |X[k]|^2/4 = \{0, 0, 4, 0\}. \text{ All periodogram power is at the Nyquist-frequency bin.}$$

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

4. Why the periodogram is inconsistent

Consistency requires the estimator variance at a fixed frequency to approach zero as N grows. For a raw periodogram of many WSS processes, including white Gaussian noise, the variance remains on the order of the squared PSD rather than vanishing.

Increasing N makes the Fourier grid finer and can improve deterministic frequency detail, but it does not create averaging across independent spectral estimates. Thus resolution and estimator variance are different issues.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

5. Bartlett design

Segment length $L = 8192/8 = 1024$ samples. Nominal bin spacing is $2048/1024 = 2$ Hz.

Averaging 8 segment periodograms reduces variance substantially (roughly like $1/K$ when the segment estimates are nearly independent). The price is frequency resolution: the full 8192-sample record would have 0.25 Hz bin spacing.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

6. Welch design trade-off

$n_{\text{perseg}} = 128$ gives nominal spacing about $2000/128 = 15.6$ Hz, so two peaks only 8 Hz apart are unlikely to be resolved. $n_{\text{perseg}} = 1024$ gives about 1.95 Hz spacing and a much narrower window main lobe, making separation more plausible.

The 128-sample choice yields many more segments for a fixed record length, hence stronger averaging and lower variance. The 1024-sample choice sacrifices averages for resolution.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

7. Window / overlap reasoning

A Hann window heavily downweights samples near each segment edge. Overlap allows those same time samples to appear nearer the center of another segment, where they receive more weight, and it increases the number of modified periodograms available for averaging.

However, overlapping segments share data, so their spectral estimates are correlated. Therefore doubling the segment count with overlap does not generally halve the variance.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

8. ML spectral features

One reasonable feature vector is log bandpower in 80–120 Hz, 220–280 Hz, plus a broadband reference band and perhaps spectral centroid. Compute all features with one fixed Welch configuration.

Fit any scaling or feature normalization on the training set only, then apply it to validation/test data. Use a pipeline such as StandardScaler + SVM or a Random Forest. Welch features can be more stable than raw FFT magnitudes because segment averaging reduces estimator variance and windowing controls leakage.

Grading guidance: full credit requires both the correct result and a brief interpretation of what it means for spectral estimation.

Weekly Quiz — Student Version

1. For a WSS process, the PSD is the _____ of the autocorrelation sequence.
2. If $S_{xx}(e^{j\omega}) = 3$ and $|H(e^{j\omega})|^2 = 0.2$ at a frequency, what is S_{yy} there?
3. Write the periodogram formula for N samples.
4. True or false: zero-padding a fixed data record creates fundamentally better frequency resolution.
5. What is the main statistical defect of the raw periodogram?
6. What does Bartlett trade to reduce variance?
7. Name the two additions Welch makes relative to Bartlett.
8. If a Welch PSD is too noisy, name one change that can reduce variance and one cost.

Weekly Quiz — Solutions

1. DTFT
2. 0.6
3. $I(\omega) = (1/N) |\sum x[n] e^{-j\omega n}|^2$
4. False
5. Its variance does not generally go to zero with N; it is inconsistent.
6. Frequency resolution / segment length.
7. Tapering windows and overlapping segments.
8. Use shorter segments / more averages or suitable overlap; cost is reduced frequency resolution or increased correlation/computation.

Instructor Notes / Differentiation

First-year emphasis: interpret plots, distinguish PSD from a single estimate, and explain the resolution–variance trade-off verbally. Senior emphasis: derive scaling, reason about window behavior, and justify estimator choices quantitatively.

Strong exit-ticket answers should identify the mechanism and the trade-off, not merely name a method. Example: “Use shorter Welch segments to gain more averages and reduce variance, but the shorter segments broaden spectral features and reduce frequency resolution.”